STUDENT COPY — NO SOLVED ANSWERS

Vyom · Is hafte ka print pack

25 Jan – 31 Jan 2027 · Monday–Sunday · NDPS Class 9

Naam: Vyom | Week tasks: 35 | Scheduled tests: 6

Sunday print handover: Sunday, 24 January 2027. Agle Monday se is pack ke daily pages follow karo. Roz tasks tick karo; tests par naam/date aur answers likho. Answer key parent ke paas rahegi.

Din / date	Kaam	Tests / focus
Monday, 25 January 2027	6 tasks	Sab Daily test (10 marks)
Tuesday, 26 January 2027	3 tasks	Phase 3 · Full revision round
Wednesday, 27 January 2027	6 tasks	Sab Daily test (10 marks)
Thursday, 28 January 2027	6 tasks	Sab Daily test (10 marks)
Friday, 29 January 2027	6 tasks	Sab Daily test (10 marks)
Saturday, 30 January 2027	6 tasks	Sab Daily test (10 marks)
Sunday, 31 January 2027	2 tasks	Science Full paper (3 hr) (80 marks)

Schedule v4 · 2026-10-12 · Sanskrit coverage. Daily blocks original shifted routine ke mutabik hain; asli weekday date har page par di hai.

School ki subject-wise exam date sheet/paper pattern abhi confirm karna baaki hai. Exam-period sample-section tasks ko official school papers mat samjhein.

Monday, 25 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sanskrit**	Revision Sanskrit full-course revision — Path 4 · नीतिवचनानि · Path 5 · दानवीरः कर्णः · Path 6: har topic ka book-band oral + written recall, ek mixed exercise, phir source PDFs se check.
☐	2:45–3:30	**Maths**	Revision Pehle homework, phir: Ch 5 · I'm Up and Down, and Round and Round (Circles) — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Maths 5.1–5.2 Circle basics · AI Self-Management · English Unit 3
☐	4:00–5:00	**Science**	Revision Full revision: Ch 5 · Exploring Mixtures and their Separation — notes + MCQ + case-based + long Q
☐	5:00–6:00	**SST**	Revision Full revision: Part 1 Ch 5 · State and Society up to 1000 CE — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 25 Jan

Maths 5.1–5.2 Circle basics · AI Self-Management · English Unit 3

Test ID: D2027-01-25 · Monday, 25 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. A circle has rotational symmetry about its _____ through any angle. **[1]**

Answer space · D2027-01-25 · Q1

2. How many lines of reflection symmetry does a circle have? **[1]**

Answer space · D2027-01-25 · Q2

3. How many circles can pass through two given points? Where do their centres lie? **[1]**

Answer space · D2027-01-25 · Q3

4. Write two signs of stress in a student and two healthy ways to manage it. **[2]**

Answer space · D2027-01-25 · Q4

5. What is self-awareness? **[1]**

Answer space · D2027-01-25 · Q5

6. Write one sign of a self-confident person. **[1]**

Answer space · D2027-01-25 · Q6

7. In “Winds of Change”, what is a *pankha*? **[1]**

Answer space · D2027-01-25 · Q7

8. Why did pankhas from different regions of India look different from one another? **[1]**

Answer space · D2027-01-25 · Q8

9. In the poem “Canvas of Soil”, what is compared to brushstrokes in “brushstrokes of seeds”? **[1]**

Answer space · D2027-01-25 · Q9

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Tuesday, 26 January 2027

Phase 3 · Full revision round · Republic Day

	Samay	Subject	Aaj kya karna hai
☐	9:30–11:00	**Hindi**	Revision Hindi: 3 paath revision + vyakaran mix
☐	11:15–12:30	**English**	Revision English: 2 units + writing task timed
☐	4:00–4:45	**Sab**	Correction Mistakes notebook dobara

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Wednesday, 27 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Hindi**	Revision Pehle homework, phir: पाठ 5 · आखिरी चट्टान तक + पाठ 6 · रीढ़ की हड्डी — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Hindi पाठ 5 · SST Ch 5 State & Society · Science Ch 5 Mixtures
☐	4:00–5:00	**Maths**	Revision Full revision: Ch 6 · Measuring Space: Perimeter and Area — notes + MCQ + case-based + long Q
☐	5:00–6:00	**Science**	Revision Full revision: Ch 6 · How Forces Affect Motion — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 27 Jan

Hindi पाठ 5 · SST Ch 5 State & Society · Science Ch 5 Mixtures

Test ID: D2027-01-27 · Wednesday, 27 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. इस पाठ के लेखक कौन हैं? **[1]**

Answer space · D2027-01-27 · Q1

2. कन्याकुमारी को 'आखिरी चट्टान' क्यों कहा गया है? **[2]**

Answer space · D2027-01-27 · Q2

3. How did trade and guilds help the economy of early India? **[3]**

Answer space · D2027-01-27 · Q3

4. How will you separate a mixture of salt and sand? **[2]**

Answer space · D2027-01-27 · Q4

5. Find the concentration (mass by mass %) of a solution with 20 g of salt in 180 g of water. **[2]**

Answer space · D2027-01-27 · Q5

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Thursday, 28 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**SST**	Revision Pehle homework, phir: Part 1 Ch 6 · Democracy — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): SST Ch 6 Democracy · Science Ch 6 Forces · Maths Ch 6 Perimeter & Area
☐	4:00–5:00	**English**	Revision Full revision: Unit 5 · The World of Limitless Possibilities + Unit 6 · Twin Melodies — notes + MCQ + case-based + long Q
☐	5:00–6:00	**AI**	Revision Full revision: Part A · Entrepreneurial Skills-I + Part A · Green Skills-I — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 28 Jan

SST Ch 6 Democracy · Science Ch 6 Forces · Maths Ch 6 Perimeter & Area

Test ID: D2027-01-28 · Thursday, 28 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. How did ancient Indian traditions contain democratic ideas? Give three points. **[3]**

Answer space · D2027-01-28 · Q1

2. Two forces of 8 N and 5 N act on a body in opposite directions. Find the net force and its direction. Is the body in equilibrium? **[2]**

Answer space · D2027-01-28 · Q2

3. Why is it easier to push an empty trolley than a loaded one? **[2]**

Answer space · D2027-01-28 · Q3

4. A circular park of radius 21 m has a 3.5 m wide path outside it. Find the area of the path. **[3]**

Answer space · D2027-01-28 · Q4

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Friday, 29 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Maths**	Revision Pehle homework, phir: Ch 7 · The Mathematics of Maybe: Probability — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Maths Ch 7 Probability · AI Entrepreneurial Skills · English Unit 5
☐	4:00–5:00	**Science**	Revision Full revision: Ch 7 · Work, Energy, and Simple Machines — notes + MCQ + case-based + long Q
☐	5:00–6:00	**SST**	Revision Full revision: Part 1 Ch 7 · Elections — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 29 Jan

Maths Ch 7 Probability · AI Entrepreneurial Skills · English Unit 5

Test ID: D2027-01-29 · Friday, 29 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. Two dice are thrown. Find the probability that the sum is (i) 7 (ii) 12 (iii) less than 4. **[3]**

Answer space · D2027-01-29 · Q1

2. A box has 3 red and 2 blue pens. Two pens are taken one after another without replacement. Using a tree diagram, find P(both blue) and P(one red and one blue). **[3]**

Answer space · D2027-01-29 · Q2

3. How do entrepreneurs help society? **[2]**

Answer space · D2027-01-29 · Q3

4. How do the Paralympics change people's attitudes? **[2]**

Answer space · D2027-01-29 · Q4

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Saturday, 30 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	6:45–7:10	**Sab**	Recall Formula / keywords notebook — ek subject
☐	2:45–3:30	**Hindi**	Revision Pehle homework, phir: पाठ 7 · मैं और मेरा देश + पाठ 8 · पद (रैदास) — notes + MCQ + case-based + long Q
☐	3:30–4:00	**Sab**	Test Daily test (10 marks, 30 min): Hindi पाठ 7 मैं और मेरा देश · SST Ch 7 Elections · Science Ch 7 Work & Energy
☐	4:00–5:00	**Maths**	Revision Full revision: Ch 8 · Predicting What Comes Next: Sequences & Progressions — notes + MCQ + case-based + long Q
☐	5:00–6:00	**Science**	Revision Full revision: Ch 8 · Journey Inside the Atom — notes + MCQ + case-based + long Q
☐	6:00–6:20	**Sab**	Recall Aaj ke 2 chapters ke 5 keywords bolke

Ho gaye: ___ / 6 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Daily Test · 30 Jan

Hindi पाठ 7 मैं और मेरा देश · SST Ch 7 Elections · Science Ch 7 Work & Energy

Test ID: D2027-01-30 · Saturday, 30 January 2027

Naam: _____ Samay: 30 min Marks: ___ / 10

Slot: 3:30–4:00

Book band. Har answer mein steps/reason likho. Calculator allowed nahi hai.

Daily questions

1. लेखक ने 'शक्तिबोध' और 'सौंदर्यबोध' से क्या समझाया है? **[2]**

Answer space · D2027-01-30 · Q1

2. लेखक को अपनी स्थिति 'हीन' क्यों लगी? **[2]**

Answer space · D2027-01-30 · Q2

3. What is the Model Code of Conduct? **[2]**

Answer space · D2027-01-30 · Q3

4. Give two differences between national and state parties. **[2]**

Answer space · D2027-01-30 · Q4

5. Find the kinetic energy of a 50 kg boy running at 4 m s^{-1}. **[2]**

Answer space · D2027-01-30 · Q5

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Sunday, 31 January 2027

Phase 3 · Full revision round

	Samay	Subject	Aaj kya karna hai
☐	9:00–12:00	**Science**	Test Science full paper — school pattern, 3 hr timed
☐	Shaam (60 min)	**Science**	Correction Checking + har galti ka reason; Mistakes notebook

Ho gaye: ___ / 2 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Science · Full Paper 1 (Annual pattern) · 31 Jan

Class 9 · Exploration — poora syllabus (Ch 1–13)

Test ID: T2027-01-31 · Sunday, 31 January 2027

Naam: _____ Samay: 180 min Marks: ___ / 80

Slot: 9:00–12:00

Paper mein 39 prashn aur 5 section hain. Saare prashn compulsory hain; Section D mein internal choice hai.
Section A: 1 mark · B: 2 marks · C: 3 marks · D: 5 marks · E: case-based 4 marks.
Numericals mein formula, substitution aur unit likho; g = 10 m s^{–2}. Diagrams labelled hon.

Section A · MCQ + Assertion-Reason · 1 mark each

1. Which step of the scientific method comes immediately after making a hypothesis? **[1]**

(a) Observation (b) Testing by experiment
(c) Conclusion (d) Asking a question

Answer space · T2027-01-31 · Q1

2. Ribosomes are the site of: **[1]**

(a) respiration (b) photosynthesis
(c) protein synthesis (d) storage of food

Answer space · T2027-01-31 · Q2

3. The tissue that joins muscle to bone is: **[1]**

(a) ligament (b) tendon
(c) cartilage (d) areolar tissue

Answer space · T2027-01-31 · Q3

4. The area under a velocity–time graph gives: **[1]**

(a) acceleration (b) speed
(c) displacement (d) force

Answer space · T2027-01-31 · Q4

5. Which of the following shows the Tyndall effect? **[1]**

(a) Salt solution (b) Sugar solution

(c) Milk (d) Copper sulfate solution

Answer space · T2027-01-31 · Q5

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6. A 5 kg body has an acceleration of 2 m s^{-2}. The net force on it is: **[1]**

(a) 2.5 N (b) 7 N

(c) 10 N (d) 3 N

Answer space · T2027-01-31 · Q6

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7. The kinetic energy of a 2 kg ball moving at 3 m s^{-1} is: **[1]**

(a) 6 J (b) 9 J

(c) 18 J (d) 3 J

Answer space · T2027-01-31 · Q7

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8. The isotopes of an element have the same: **[1]**

(a) mass number (b) number of neutrons

(c) atomic number (d) nuclear mass

Answer space · T2027-01-31 · Q8

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9. The valency of nitrogen in NH_3 is: **[1]**

(a) 1 (b) 2

(c) 3 (d) 5

Answer space · T2027-01-31 · Q9

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10. The formula unit mass of NaCl (Na = 23, Cl = 35.5) is: **[1]**

(a) 58.5 u (b) 23 u

(c) 35.5 u (d) 46 u

Answer space · T2027-01-31 · Q10

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11. Humans can hear sounds of frequency: **[1]**

(a) below 20 Hz (b) 20 Hz to 20 kHz

(c) above 20 kHz (d) only above 1 kHz

Answer space · T2027-01-31 · Q11

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12. Persistence of sound in a hall due to repeated reflections is called: **[1]**

(a) echo
(b) reverberation
(c) resonance
(d) refraction

Answer space · T2027-01-31 · Q12

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13. In flowering plants, the ovule develops into the: **[1]**

(a) fruit
(b) seed
(c) flower
(d) pollen

Answer space · T2027-01-31 · Q13

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14. Organisms with prokaryotic cells are placed in kingdom: **[1]**

(a) Protista
(b) Fungi
(c) Monera
(d) Plantae

Answer space · T2027-01-31 · Q14

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15. Fraction of sunlight reflected by a surface is its: **[1]**

(a) albedo
(b) humidity
(c) latitude
(d) insolation

Answer space · T2027-01-31 · Q15

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16. Which process removes carbon dioxide from the atmosphere? **[1]**

(a) Respiration
(b) Burning of fuels
(c) Photosynthesis
(d) Decomposition

Answer space · T2027-01-31 · Q16

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17. **A:** A person sitting in a moving bus falls backward when the bus starts suddenly. **[1]**
R: The lower part of the body moves with the bus, but the upper part tends to remain at rest due to inertia.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Answer space · T2027-01-31 · Q17

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18. **A:** Work done by gravity on a satellite moving in a circular orbit is zero. **[1]**
R: Gravitational force on the satellite is perpendicular to its displacement at every point.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2027-01-31 · Q18

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19. **A:** Sound travels faster in air than in steel. **[1]**
R: Particles are closer and interact more strongly in solids.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2027-01-31 · Q19

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20. **A:** Burning large amounts of fossil fuels adds to global warming. **[1]**
R: Carbon dioxide is a greenhouse gas that traps heat re-radiated by the Earth.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2027-01-31 · Q20

Section B · Very Short Answer · 2 marks each

21. Write two differences between plant and animal tissues. **[2]**

Answer space · T2027-01-31 · Q21

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22. A car travels 100 m north and then 100 m back south in 40 s. Find its average speed and average velocity. **[2]**

Answer space · T2027-01-31 · Q22

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23. How will you separate (i) iron filings from sand (ii) cream from milk? **[2]**

Answer space · T2027-01-31 · Q23

24. An atom has mass number 27 and 14 neutrons. Find its atomic number and electronic configuration. **[2]**

Answer space · T2027-01-31 · Q24

25. Give two reasons why biodiversity is important for an ecosystem. **[2]**

Answer space · T2027-01-31 · Q25

26. How is a land breeze formed at night? **[2]**

Answer space · T2027-01-31 · Q26

Section C · Short Answer · 3 marks each

27. Draw a labelled diagram of an animal cell showing the nucleus, mitochondria and cell membrane. Write one function of each. **[3]**

Answer space · T2027-01-31 · Q27

28. A bus starting from rest moves with uniform acceleration of 0.5 m s^{-2} for 1 minute. Find (i) the speed acquired (ii) the distance travelled. **[3]**

Answer space · T2027-01-31 · Q28

29. A solution has 40 g of common salt in 320 g of water. Find the mass by mass percentage. Classify as true solution, suspension or colloid: blood, brass, chalk in water. **[3]**

Answer space · T2027-01-31 · Q29

30. A bullet of mass 20 g is fired from a gun of mass 4 kg with a velocity of 400 m s^{-1}. Find the recoil velocity of the gun. **[3]**

Answer space · T2027-01-31 · Q30

31. Calculate the molecular masses of C_2H_5OH and $CaCO_3$. (C = 12, H = 1, O = 16, Ca = 40) **[3]**

Answer space · T2027-01-31 · Q31

32. A person standing 170 m from a cliff claps. After how much time will he hear the echo (speed of sound 340 m s^{-1})? Write two uses of ultrasound. **[3]**

Answer space · T2027-01-31 · Q32

33. Describe binary fission in Amoeba. How is it different from multiple fission? **[3]**

Answer space · T2027-01-31 · Q33

Section D · Long Answer · 5 marks each

34. State the three laws of motion with one example each. Show that the second law gives F = ma. **[5]**

ATHVA / OR

Define work, energy and power with their SI units. A 50 kg girl climbs 4 m in 8 s. Find the work done and power. Explain how a pulley and an inclined plane make work easier.

Answer space · T2027-01-31 · Q34

35. Explain how ionic and covalent bonds are formed, with one example each (electron-dot diagrams). Write two properties of each type of compound. **[5]**

Answer space · T2027-01-31 · Q35

36. Describe the five-kingdom classification. Give two features and one example of each kingdom. **[5]**

ATHVA / OR

Describe the carbon cycle and the nitrogen cycle. How do human activities disturb the carbon cycle?

Answer space · T2027-01-31 · Q36

Section E · Case-Based · 4 marks each

37. **[4]**

> A ball is dropped from the top of a 45 m tall building. Ignore air resistance; g = 10 m s^{-2}.

(i) How long does it take to reach the ground? [1]
(ii) With what speed does it hit the ground? [1]
(iii) Sketch its velocity–time graph and state what the slope of the graph represents. What is the distance covered in the first 2 s? [2]

Answer space · T2027-01-31 · Q37

38. **[4]**

> Element X has 12 protons and 12 neutrons. Element Y has atomic number 17 and mass number 35. Another atom Z has atomic number 17 and mass number 37.

(i) Write the electronic configuration of X. [1]
(ii) How are Y and Z related? [1]
(iii) What ions do X and Y form? Write the formula of the compound formed between X and Y and its formula unit mass (X = 24 u, Y = 35.5 u). [2]

Answer space · T2027-01-31 · Q38

39. **[4]**

> Night-blooming flowers like Raat ki Rani are white or pale and strongly scented, while many day flowers are bright coloured. Bees, butterflies, moths and bats visit flowers for nectar.

(i) What is pollination? [1]
(ii) Why are night-blooming flowers usually white and fragrant? [1]
(iii) Why is cross-pollination useful for a plant species? What happens after pollination for a seed to form? [2]

Answer space · T2027-01-31 · Q39

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Extra answer sheet · 25 Jan – 31 Jan

Naam: _____ Date: _____ Subject: _____

Test ID / question number: _____

Diagram / graph aur Sunday review

Test ID / question number: _____



Is hafte ki 3 galtiyan aur sudhaar:

Week tasks: ___ / 35 Parent sign: _____

Agla hafta: sabse zaroori weak topic / pending kaam: